Trigonometric Functions and Circular Measure

Section 1: Radian Measure, Arc Length and Sectors

1 A circle of radius $8\ \text{cm}$ has a central sector with angle $\theta = \frac{3\pi}{4}\ \text{rad}$. Find the exact arc length of the sector.

a. $6\pi\ \text{cm}$

b. $12\pi\ \text{cm}$

c. $4\pi\ \text{cm}$

d. $24\pi\ \text{cm}$

2 Find the area of a sector of radius $6\ \text{cm}$ having perimeter $22\ \text{cm}$.

a. $30\ \text{cm}^2$

b. $36\ \text{cm}^2$

c. $24\ \text{cm}^2$

d. $18\ \text{cm}^2$

3 A chord of length $10\ \text{cm}$ subtends an angle of 2θ at the center of a circle of radius $13\ \text{cm}$. Find $\sin\theta$.

a. $\frac{5}{13}$

b. $\frac{12}{13}$

c. $\frac{10}{13}$

d. $\frac{5}{12}$

Section 2: Trigonometric Identities and Ratios

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4 Simplify the trigonometric expression $\frac{\sin\theta}{\cos\theta} + \frac{\cos\theta}{\sin\theta}$.

a. $\sec\theta\csc\theta$

b. $\tan\theta$

c. 1

d. $\sec^2\theta$

5 Given that $\tan\theta = -\frac{3}{4}$ and θ lies in the second quadrant, find the value of $\cos\theta$.

a. $-\frac{4}{5}$

b. $\frac{4}{5}$

c. $-\frac{3}{5}$

d. $\frac{3}{5}$

6 Prove the identity $\sec^2\theta - \tan^2\theta = 1$. Which of the following equals $(\sec\theta - \tan\theta)(\sec\theta + \tan\theta)$?

a. 1

b. $\sec^2\theta$

c. $2\tan^2\theta$

d. 0

Section 3: Solving Trigonometric Equations

7 Solve $2\cos(2x) = 1$ for $0 \le x \le \pi$.

a. $x = \frac{\pi}{6}$ and $x = \frac{5\pi}{6}$

b. $x = \frac{\pi}{3}$ and $x = \frac{2\pi}{3}$

c. $x = \frac{\pi}{6}$ only

d. $x = \frac{\pi}{4}$ and $x = \frac{3\pi}{4}$

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8 Find all solutions of $3\sin x = 2\cos^2 x$ in the interval $0^\circ \le x \le 360^\circ$.

a. 30° and 150°

b. 30° and 210°

c. 60° and 120°

d. 45° and 135°

9 Solve the equation $\tan(x + 45^\circ) = 1$ for $0^\circ \le x \le 360^\circ$.

a. $x = 0^\circ, 180^\circ, 360^\circ$

b. $x = 45^\circ, 225^\circ$

c. $x = 90^\circ, 270^\circ$

d. $x = 0^\circ$ only

Section 4: Double Angle Formulae and Graph Amplitude

10 If $\sin\theta = \frac{3}{5}$ and θ is acute, find the exact value of $\sin(2\theta)$.

a. $\frac{24}{25}$

b. $\frac{12}{25}$

c. $\frac{7}{25}$

d. $\frac{6}{5}$

11 State the amplitude and period of the function $y = 3\sin(4x) - 2$.

a. Amplitude $= 3$, Period $= \frac{\pi}{2}$

b. Amplitude $= 3$, Period $= \frac{\pi}{4}$

c. Amplitude $= 1$, Period $= 2\pi$

d. Amplitude $= 4$, Period $= \frac{\pi}{2}$

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12 Express $\cos(2\theta)$ purely in terms of $\sin\theta$.

a. $1 - 2\sin^2\theta$

b. $2\sin^2\theta - 1$

c. $1 - \sin^2\theta$

d. $\sin^2\theta - 1$

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